Causal Inference for Intervention & Service Evaluations: A Practical Handbook

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01 Dec 2025

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Welcome

This is a practical handbook on causal inference methods for evaluating policy interventions and service changes within the NHS, with a focus on leveraging observational real-world data to generate robust evidence.

The handbook was written initially as part of an NHS England PhD Internship done jointly with the NHS North Central London Integrated Care Board and University College London, and will continue to be periodically updated with revisions following feedback from academics, data scientists, and public health practitioners.

NHS England

NHS North Central London Integrated Care Board

UCL

Part I

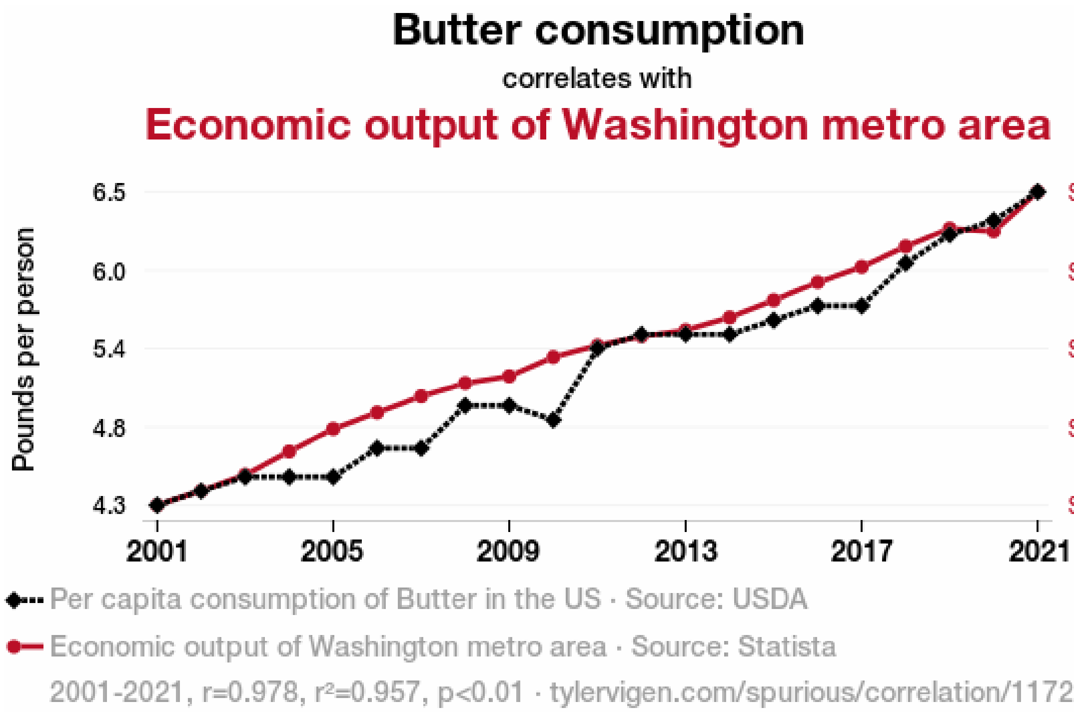
Introduction

1 Background & Motivation for Causal Inference in Service Evaluations

When performing evaluations of healthcare services or interventions in the NHS, our key evaluation question of interest is almost always causal in nature: • Has our new breast cancer screening programme reduced mortality? • Did implementing a community-based model of care divert patients away from A&E services? • Did implementing a hub and spoke model reduce clinical harm and lower patient waiting times?

Yet many of the methods currently used to evaluate interventions within the NHS are often descriptive in nature. Dashboards are frequently used in this manner, where any changes in an outcome of interest are fully attributed to a single change in implementation in services, without considering whether there are alternative explanations and/or other confounding factors are involved that could instead explain the change instead.

To better illustrate this point, Tyler Vigen publishes great examples of spurious correlations on his webpage (https://tylervigen.com/spurious-correlations), from which I've attached an example of below Vigen (2024):


Butter consumption
correlates with
Economic output of Washington metro area
Pounds per person
6.5
6.0
5.4
4.8
4.3
$607.5T
$521.5T
$435.4T
$349.4T
$263.3T
Dollars
2001
2005
2009
2013
2017
2021
Per capita consumption of Butter in the US · Source: USDA
Economic output of Washington metro area · Source: Statista
2001-2021, r=0.978, r²=0.957, p<0.01 · tylervigen.com/spurious/correlation/1172

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30.1 About Me

Hi, my name is Jamie and I'm an MRC funded PhD student at UCL, focusing on the application of causal inference methods within Epidemiology, Data Science, and Health Economics.

Before my PhD, I previously graduated with a BSc in Population Health Sciences (Data Science) from UCL, and an NIHR funded MSc in Epidemiology from the London School of Hygiene and Tropical Medicine.

Prior to my PhD, my interest in researching the application of causal inference methods in observational data stemmed from my varied experience working across multiple areas of public health research, including: • Performing secondary analysis on a nutritional labelling RCT at the NIHR Obesity Policy Research Unit • Supporting cancer trials as a Clinical Trials Assistant in the MRC Clinical Trials Unit • Identifying barriers to STI testing at the NIHR Health Protection Research Unit in Blood Borne and Sexually Transmitted Infections • Quantifying the effects of acute and long-term exposure of air pollutants on respiratory related hospital admissions and related outcomes at UCL • Evaluating the risk of long-term adverse events in cancer survivors using electronic health record data at LSHTM

Having worked with both randomized trials and observational data, focusing my research on causal inference in real-world data felt like a natural next step.

If you have any questions do feel free to reach out to me via the following channels: UCL Email: jamie.wong [at] ucl.ac.uk NHS Email: laichunjamie.wong [at] nhs.net LinkedIn: https://www.linkedin.com/in/jamiewlc/ GitHub: https://github.com/jamiewlc

30.2 Acknowledgements

This handbook was written during my PhD placement with NHS England and the North Central London Integrated Care Board as part of the NHS England Data science PhD internship programme, supervised by Jonathan Pearson (NHSE) and Emily Baldwin (NCL ICB).

References

Vigen, Tyler. 2024. "Tyler Vigen Spurious Correlation #1,172: Butter Consumption Correlates with Economic Output of Washington Metro Area." TylerVigen.com. January 2024. https://tylervigen.com/spurious/correlation/1172_butter-consumption_correlates-with_economic-output-of-washington-metro-area.